Hager Flow
ModbusTCPServer Table



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Version

Version	FR2 Release	Date	Changes
0.1	FR2_2024_03	17/06/2024	First version

SlaveID independant

Device Identification

Description	Unit	Address DEC	Address HEX	Lenght (word)	Data Type	Read/Write	XEM470	Comment
Vendor Name		0	0	16	STRING	Read only	"Hager"	
Product Code		16	10	16	STRING	Read only	"XEM470"	
Vendor URL		32	20	16	STRING	Read only	“http://www.hager.com”	
Product code (Serial number)		48	30	16	STRING	Read only	3250615600516	
SW version		64	40	16	STRING	Read only	FR2_2023_054	

General Information

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
Max Current Phase	A	272	110	1	U16	Read only	0…2500	Main Fuse
Max Current Phase Derated (80%)	A	273	111	1	U16	Read only	0…2500	Main Fuse derated
Max Power Phase	W	274	112	2	U32	Read only	0…9,999,999	Main Fuse
Max Power Phase Derated (80%)	W	276	114	2	U32	Read only	0…9,999,999	Main Fuse derated
Max Power	W	278	116	2	U32	Read only	0…9,999,999	Max Installation power from grid

Max Power Derated (80%)	W	280	118	2	U32	Read only	0…9,999,999	Max Installation power from grid derated
EVCS								**EVCS**
EVCS Number		306	132	1	U16	Read only	0…7	Number of Connected EVSE
EVCS list		307	0133	13	BITSTRING (0…255)	Read only	0..255	Positon of connected EVSE : bit 0 : EVCS 1 bit 1 : EVCS 2 … bit 7 : EVCS 8
EMS Configuration and status								
Phase Number		512	200	1	U16	Read only	1..3	One phase or Three phases
Sun Power priority						Read only	0...1... 2	bit 0 :1 = Battery First / 0= Car first bit 1: 1= allow battery discharge into car / 0 = forbid battery discharge into car
Blackout Active		514	202	1	U16	Read only		NOT_POSSIBLE = 0x00, ACTIVE = 0x01, NOT_ACTIVE = 0x02, NOT_AVAILABLE = 0x03, SWITCH_IN_ISLAND_STATE = 0x04
SGReady Brain disable		515	203	1	U16	Read only	0/1	1 = SGReady Brain disabled 0 = SGReady Brain enabled
SGReady Power	W	516	204	2	U32	Read only	0..9'999'999	Total nominal power controlled by SG-Ready in Watts

Slave ID identity

Description	Unit	Address DEC	Address HEX	Lenght (word)	Data Type	Read/Write	XEM470	comment
SlaveID type		4096	1000	1	U16	Read only		0: Main system (EMS) 1: EVSE 2: Storage 3: Power Meter 4: SG-Ready
Slave ID index by type		4097	1001	1	U16	Read only		0..99

Slave ID 0

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
Root/Main Current (ΣL)	A	4098	1002	1	U16	Read only	0…7500	
Root/Main Current (L1)	A	4099	1003	1	U16	Read only	0…2500	
Root/Main Current (L2)	A	4100	1004	1	U16	Read only	0…2500	

Root/Main Current (L3)	A	4101	1005	1	U16	Read only	0…2500	
Root/Main Power (ΣL)	W	4102	1006	2	U32	Read only	0…9,999,999	
Root/Main Power (L1)	W	4104	1008	2	U32	Read only	0…9,999,999	
Root/Main Power (L2)	W	4106	100A	2	U32	Read only	0…9,999,999	
Root/Main Power (L3)	W	4108	100C	2	U32	Read only	0…9,999,999	
All EVCS Current (ΣL)	A	4110	100E	1	U16	Read only	0…2500	
All EVCS Power (ΣL)	W	4114	1012	2	U32	Read only		
All PV Current (ΣL)	A	4122	101A	1	U16	Read only	0…2500	
All PV Power (ΣL)	W	4126	101E	2	U32	Read only		
All Battery Current (ΣL)	A	4134	1026	1	U16	Read only	0…2500	
All Battery Power (ΣL)	W	4138	102A	2	I32	Read only		
Battery State of Charge	%	4146	1032	1	U16	Read only	0…100	
All HouseConsumption Current (ΣL)	A	4147	1033	1	U16	Read only	0…2500	
All HouseConsumption Power (ΣL)	W	4151	1037	2	I32	Read only		
Autarky (last hour)	%	4224	1080	1	U16	Read only	0…100	
Self Consumption (last hour)	%	4225	1081	1	U16	Read only	0…100	
MarginToMaxPowerTotal	W	4240	1090	2	I32	Read only		
MarginToMaxPowerL1	W	4242	1092	2	I32	Read only		
MarginToMaxPowerL2	W	4244	1094	2	I32	Read only		
MarginToMaxPowerL3	W	4246	1096	2	I32	Read only		

Slave ID 1-8 : EVSE

Generic WallBox information

Identity information

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
EVCS Number		4097	1001	1	U16	Read only	1…8	
EVCS Type		4098	1002	1	U16	Read only	0/1	Wallbox type: 0: virtual 1: E3DC wallbox 2: Easy Connect wallbox 4: wallbox RSCP group (farming) 5: Edison Connect wallbox 6: Multi Connect wallbox 7: Eebus Wallbox
EVCS Name		4099	1003	50	STRING	Read only	Friendly name set by User	

EVCS Serial		4149	1035	16	STRING	Read only	Suid	
EVCS Firmware version		4165	1045	32	STRING	Read only	Firmware version	

General Status and configuration

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
EVCS State		4608	1200	1	U16	Read only		Returns WTC mode byte, functions as a bitmask: Bit 0: normal mode Bit 1: default mode (CRC wrong parameter value range) Bit 2: sun mode active Bit 3: temperature derating active Bit 4: Schuko active Bit 5: CRC wrong meter value range Bit 6: limited operation (not all modes active) Bit 7: empty
EVCS Energy All	Wh	4609	1201	2	U32	Read only	0…9,999,999	EVCS Energy index
EVCS Solar Energy	Wh	4611	1203	2	U32	Read only	0…9,999,999	EVCS Solar Energy index
EVCS Connection State		4613	1205	1	U16	Read only	0/1	0 : EVCS disconnected 1 : EVCS connected

EVCS Working State		4614	1206	1	U16	Read only	0/1	0 : EVCS not working properly 1 : EVCS working normally
EVCS Service State		4615	1207	1	U16	Read only	0/1	0 : no upgrade ongoing 1: firmware upgrade ongoing
EVCS Power Meter Connection State		4616	1208	1	U16	Read only	0/1	0 : EVCS power meter disconnected 1 : EVCS power meter connected
EVCS Power Meter Working State		4617	1209	1	U16	Read only	0/1	0 : EVCS power meter not working properly 1 : EVCS power meter working normally
EVCS Power Meter Service State		4618	120A	1	U16	Read only	0/1	0 : no upgrade ongoing 1: firmware upgrade ongoing
EVCS Availability		4619	120B	1	U16	Read only	0/1	0 : Unavailable (charge is forbidden) 1 : Available (charge might be authorized)
EVCS Station Enabled		4620	120C	1	U16	Read only		0 : Disabled 1 :Enabled
EVCS Max Charging Power	W	4621	120D	2	U32	Read only	0…9,999,999	In total for all phases. This value might be changed when charging session is started.
EVCS Max Charging Power per Phase	W	4623	120F	2	U32	Read only		Per phase

EVCS Max Charging Current	A	4625	1211	1	U16	Read only	6…32	EVCS max allowed charging current
EVCS Min Charging Power	W	4626	1212	2	U32	Read only		In total for all phases. This value might be changed when charging session is started.
EVCS Min Charging Power per Phase	W	4628	1214	2	U32	Read only		Per phase This value might be changed when charging session is started. (EV41 for example)
EVCS Min Charging Current	A	4630	1216	1	U16	Read only	6/32	EVCS min allowed charging current
EVCS Boost Mode		4631	1217	1	U16	Read / Write	0/1	0 : Boost mode inactive (if solar mode enable wait PV to start charge) 1 : Boost mode active (charge at full power even in solar mode) This mode is reset at the end of the next EVCS charging session
EVCS Phase Management		4632	1218	1	U16	Read only	1/3/4	1=mono 3=tri 4=auto

EVCS Charge Strategy		4633	1219	1	U16	Read only	1..4	0 = Boost (Immediate) 1 : Delayed 2: disabled
EVCS Charge Strategy daytime		4634	121A	64	U16	Read only	0..86'400; 0..86'400; 0..86'400; 0..86'400; 0..86'400; 0..86'400; 0..86'400;	departure time in (in seconds from midnight) for: SUNDAY; MONDAY; TUESDAY; WEDNESDAY; THURSDAY; FRIDAY; SATURDAY;
EVCS Charge Strategy Minimum Energy Value		4698	125A	64	U16	Read only	0..9999; 0..9999; 0..9999; 0..9999; 0..9999; 0..9999; 0..9999;	Minimal charging energy before depature time in Wh for: SUNDAY; MONDAY; TUESDAY; WEDNESDAY; THURSDAY; FRIDAY; SATURDAY;

Charging Session Information

This data are only relevant during a charging session.

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment

EVCS State		4864	1300	1	U16	Read only	WTC mode	Returns WTC mode byte, functions as a bitmask: Bit 0: normal mode Bit 1: default mode (CRC wrong parameter value range) Bit 2: sun mode active Bit 3: temperature derating active Bit 4: Schuko active Bit 5: CRC wrong meter value range Bit 6: limited operation (not all modes active) Bit 7: empty
Charging Session ID		4865	1301	64	String	Read only		
Charging Session Badge ID		4929	1341	16	STRING	Read only		
Status		4945	1351	4	U16	Read only	ALG Data	Refer to ALG Data
Active Phase		4949	1355	1	U16	Read only		Refer to Phase mapping Enum
Energy Session	Wh	4950	1356	2	U32	Read only	0…9,999,999	Energy meter since start of current charging session, restart at 0 Wh at each charging session
Grid Energy Session	Wh	4952	1358	2	U32	Read only	0…9,999,999	Energy meter since start of current charging session, restart at 0 Wh at each charging session
Solar Energy Session	Wh	4954	135A	2	U32	Read only	0…9,999,999	Energy meter since start of current charging session, restart at 0 Wh at each charging session

Instantaneous Measures

These values are only relevant during a charging session.

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
Total Charging Current	A	5120	1400	1	U16	Read only	0…32	
Charging Current (L1)	A	5121	1401	1	U16	Read only	0…32	
Charging Current (L2)	A	5122	1402	1	U16	Read only	0…32	
Charging Current (L3)	A	5123	1403	1	U16	Read only	0…32	
Charging Power Total	W	5124	1404	2	U32	Read only	0…22000	
Charging Power (L1)	W	5126	1406	2	U32	Read only	0…7500	
Charging Power (L2)	W	5128	1408	2	U32	Read only	0…7500	
Charging Power (L3)	W	5130	140A	2	U32	Read only	0…7500	
Assign Power Total	W	5132	140C	2	U32	Read only	0…22000	
Assigned Power (L1)	W	5134	140E	2	U32	Read only	0…7500	
Assigned Power (L2)	W	5136	1410	2	U32	Read only	0…7500	
Assigned Power (L3)	W	5138	1412	2	U32	Read only	0…7500	

Modbus Wallbox specific

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
MAC Address		5376	1500	9	STRING	Read only	0..2	
IP address		5385	1509	8	STRING	Read only	0..255.0..255.0..255.0..255.	Wallbox IP address
Gateway		5393	1511	8	STRING	Read only	0..255.0..255.0..255.0..255.	
Sub Network mask		5401	1519	8	STRING	Read only	0..255.0..255.0..255.0..255.	
DHCP Enabled		5409	1521	1	U16	Read only	0/1	0 : DHCP client inactive (Network configuration 1 : DHCP client active (Network configuration)
EVCS RFID Enable		5411	1523	1	U16	Read only	0..2	0 : Free Access Mode 1 : Badge Access Mode
EVCS RFID reader		5412	1524	16	STRING	Read only		Returns the RFID data of the current charging session

Slave ID 20 : Storage

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
Storage Index		4097	1001	1	U16	Read only	0..255	
Storage DCDC status		4098	1002	1	bool	Read only	0/1	
Storage Bat Status		4099	1003	1	bool	Read only	0/1	
Battery State of Charge	%	4100	1004	1	U16	Read only	0…100	

Slave ID 30-37 : PowerMeter

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
PowerMeter Id		4097	1001	1	U16	Read only	0..7	
PowerMeter Name		4098	1002	50	STRING	Read only		
PowerMeter Type		4148	1034	1	U16	Read only	0..8	0 = UNDEFINED 1 = ROOT 2 = ADDITIONAL 3 = ADDITIONAL PRODUCTION 4 = ADDITIONAL CONSUMPTION 5 = FARM 6 = UNUSED 7 = WALLBOX 8 = FARM ADDITIONAL

PowerMeter Connection status		4149	1035	1	bool	Read only	0 = Not connected 1 = Connected	
PowerMeter Working status		4150	1036	1	bool	Read only	0 = Not Working 1 = Working	
PowerMeter In service status		4151	1037	1	bool	Read only	0 = Not Working 1 = Working	
PowerMeter Active phase		4152	1038	1	3-bit bitfield	Read only	bit 0 = L1 bit 1 = L2 bit 2 = L3	
PowerMeter Power (ΣL)	W	4153	1039	2	I32	Read only		
PowerMeter Power (L1)	W	4155	103B	2	I32	Read only		
PowerMeter Power (L2)	W	4157	103D	2	I32	Read only		
PowerMeter Power (L3)	W	4159	103F	2	I32	Read only		
PowerMeter Voltage (L1)	V	4161	1041	1	U16	Read only	0...400	
PowerMeter Voltage (L2)	V	4162	1042	1	U16	Read only	0...400	
PowerMeter Voltage (L3)	V	4163	1043	1	U16	Read only	0...400	

PowerMeter Energy (ΣL)	Wh	4164	1044	4	F64	Read only		
PowerMeter Energy (L1)	Wh	4168	1048	4	F64	Read only		
PowerMeter Energy(L2)	Wh	4172	104C	4	F64	Read only		
PowerMeter Energy (L3)	Wh	4176	1050	4	F64	Read only		

Slave ID 50-59 : SG-ReadyDevice

Description	Unit	Address DEC	Address HEX	Length (word)	Data Type	Read/Write	XEM470	Comment
SGReady index		4097	1001	1	U16	Read only	0..9	
SGReady Name		4098	1002	50	STRING	Read only	0..9	
SGReady active		4150	1036	1	bool	Read only	0..1	0 = Inactive 1= active
SGReady working		4151	1037	1	bool	Read only	-3..0	0 = OK -1 = hardware issue (pin service not found) -2 = reserved -3 = SGReady controller not alive

SGReady status		4152	1038	2	I32	Read only	1..4	STATE_1_BLOCK = 1, STATE_2_NORMAL = 2, STATE_3_GO = 3, STATE_4_FORCE_GO = 4

Annexes 1 : ALG Data

EXTERN_DATA byte array (length 8) contains:

Bytes		
Byte	**Type**	**Description**
1	uint8_t	pre charge [%]
2	uint8_t	number phases [0-3]
3	status bitmask	see below Byte 3
4	uint8_t	current limitation [A]
5	status bitmask	see below Byte 5
6	unused	
7	uint8_t	display info/warning/error, see below

Byte 3		
bit position	**reset**	**set**
7	mix mode active	sun mode active
6	charge enabled	charge aborted
5	EV not charging	EV charging
4	type 2 connector unlocked	type 2 connector locked
3	type 2 connector unplugged	type 2 connector plugged
2	Schuko disabled	Schuko enabled
1	Schuko unplugged	Schuko plugged
0	Schuko unlocked	Schuko locked
Byte 5		
bit position	**reset**	**set**
7	LED-ERR off	LED-ERR on (red)
6	LED-SON off	LED-SON on (yellow)

5	LED-BAT off	LED-BAT on (green)
4	Relais off, 16A 1Phase, Schuko	Relais on, 16A 1Phase, Schuko
3	Relais off, 16A 3Phases, Type 2	Relais on, 16A 3Phases, Type 2
2	Relais off, 32A 3Phases, Type 2	Relais on, 32A 3Phases, Type 2
1	EV ventilation off	EV ventilation on
0	*Unused*	

Byte 7 : diag (depracted ?)	
Value	**Description**
0x00	none
Info	
0x10	Schuko not possible, internal PowerMeter failure
0x11	Schuko not possible, type 2 > 16A
0x12	Schuko not possible, temperature > 70*C
0x13	Schuko not possible, emergency power operation
0x14	Schuko not possible, type 2 over Schuko
0x15	not possible, Schuko or type 2 plugged
0x16	Schuko not possible, externally locked
0x17	?
0x18	sun mode not possible, no external PowerMeter
0x19	external abort
0x1A	overflow
0x1B	wrong password
0x1C	password changed
Warnings	
0x20	casing temperature > 50*C, derating
0x21	internal PowerMeter not available
0x22	FRAM CRC error, default parameters

0x23	Flash CRC error
0x24	FRAM CRC error, meter values
0x25	emergency power operation active
0x26	negative power at type 2 connector
0x27	T sensor not calibrated, max current 16A
Errors	
0x40	FRAM failure, default parameters
0x41	Flash failure
0x42	CAN failure
0x43	casing temperature > 70*C, disable everything
0x44	charging error, CP level in grey area
0x45	charging error, diode failure
0x46	charging error, PP unknown
0x47	charging gear failure, PP control